

Objective

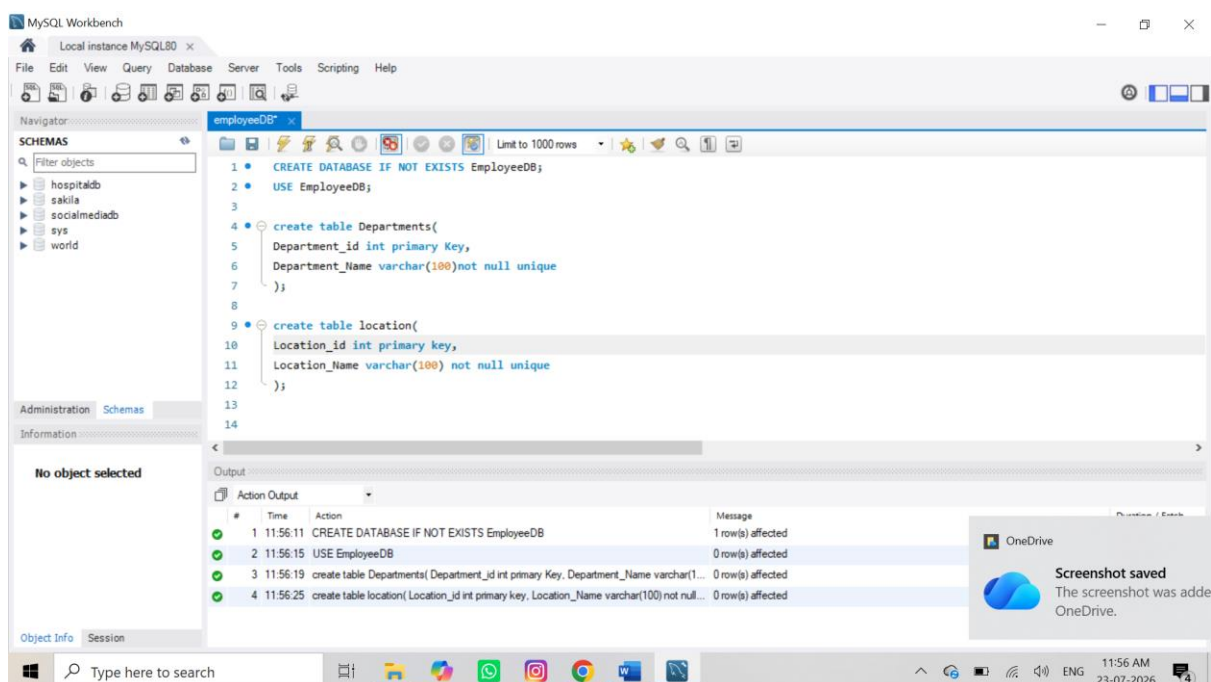
To design and manage an Employee Database using MySQL DDL(Data Definition Language) commands by creating databases, tables, and relationships with appropriate constraints, ensuring data integrity, consistency, and efficient management of employee, department, and location information within the organization.

Create Table

1. Table: Departments

Attributes:

- department_id – Primary Key
- department_name

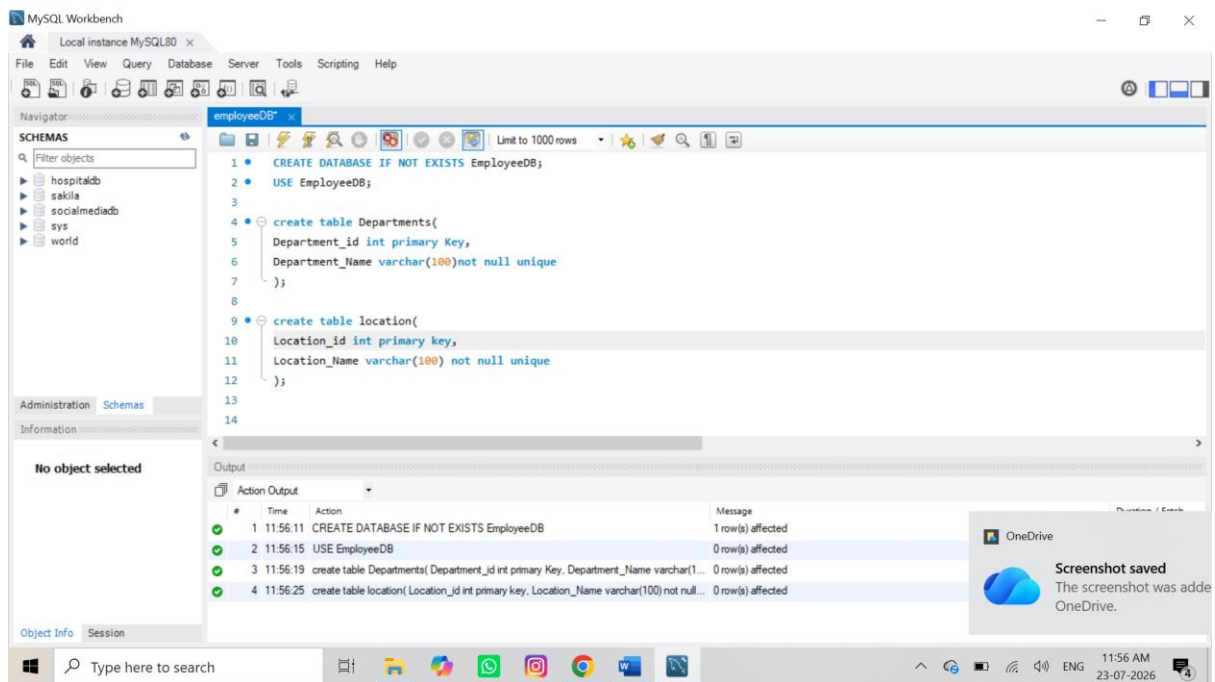


Prevents empty (NULL) values in the department name

2.Table: Location

Attributes:

- location_id – Primary Key
- location_name

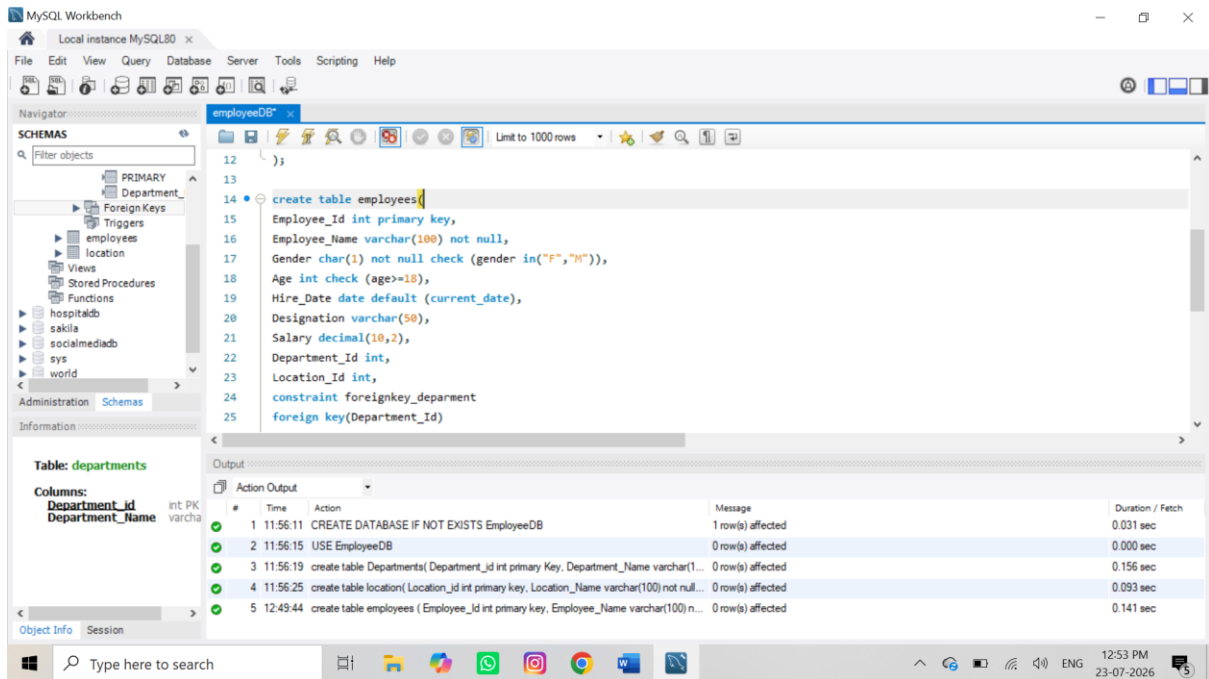


Prevents empty (NULL) values in the department name

3.Table: Employees

Attributes:

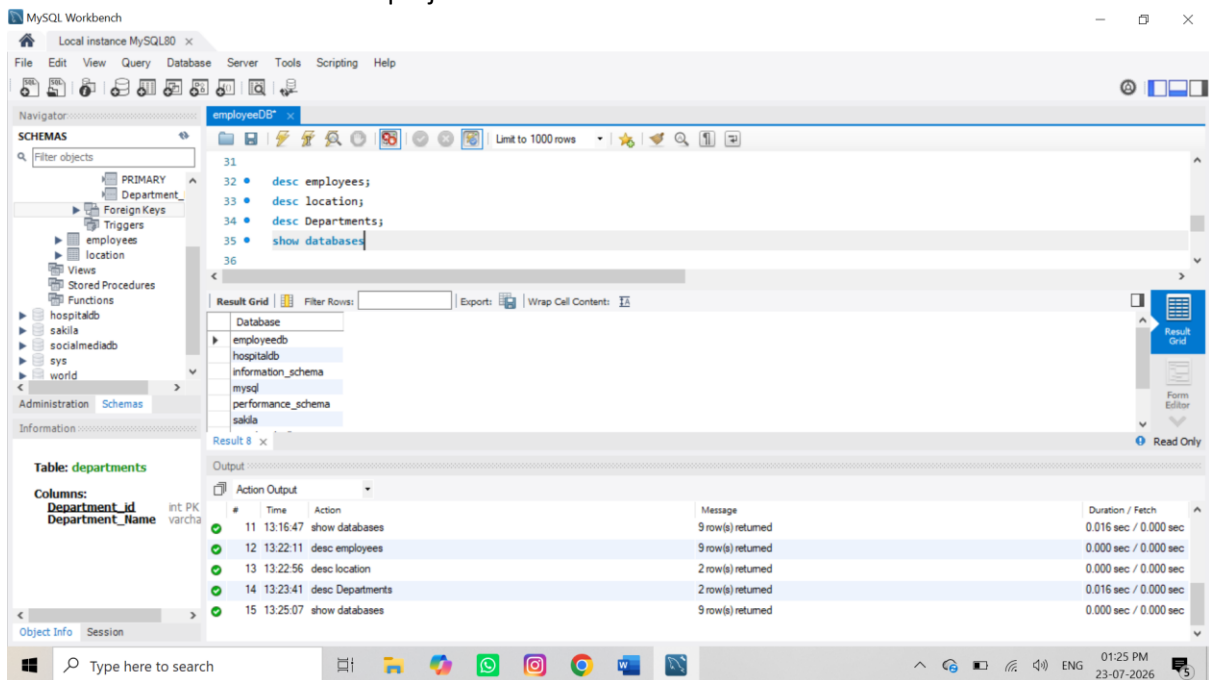
- employee_id – Primary Key
- Employee_name
- Gender
- Age
- Hire_date
- Designation
- Salary
- department_id – Foreign Key referencing department(department_id)
- location_id—Foreign Key referencing location(location_id)



DDL Commands

1. Database and Table Creation (CREATE):

- Create a Database for this project



- Create all three tables in MySQL with appropriate data types and relationships.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

PRIMARY Department

Foreign Keys

Triggers

employees

location

Views

Stored Procedures

Functions

hospitaldb

sakila

socialmediadb

sys

world

Administration Schemas

Information

Table: departments

Columns:

Department_id int PK

Department_name varchar

employeeDB

Limit to 1000 rows

30

31

32 desc employees;

33

34

35

Result Grid

Field	Type	Null	Key	Default	Extra
Employee_id	int	NO	PRI		
Employee_name	varchar(100)	NO			
Gender	char(1)	NO			
Age	int	YES			
Hire_date	date	YES		curdate()	DEFAULT_GENERATED
Designation	varchar(50)	YES			

Result 5

Output

#	Time	Action	Message	Duration / Fetch
8	13:13:09	(table)	Error Code: 1064. You have an error in your SQL syntax; check the manual that...	0.000 sec
9	13:13:25	desc employees	9 row(s) returned	0.000 sec / 0.000 sec
10	13:15:34	show tables	3 row(s) returned	0.000 sec / 0.000 sec
11	13:16:47	show databases	9 row(s) returned	0.016 sec / 0.000 sec
12	13:22:11	desc employees	9 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

01:22 PM 23-07-2026

MySQL Workbench

Local instance MySQL80

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Table: departments

Columns:

Department_id int PK

Department_name varchar

employeeDB

Limit to 1000 rows

30

31

32 desc employees;

33 desc location;

34

35

Result Grid

Field	Type	Null	Key	Default	Extra
Location_id	int	NO	PRI		
Location_name	varchar(100)	NO	UNI		

Result 6

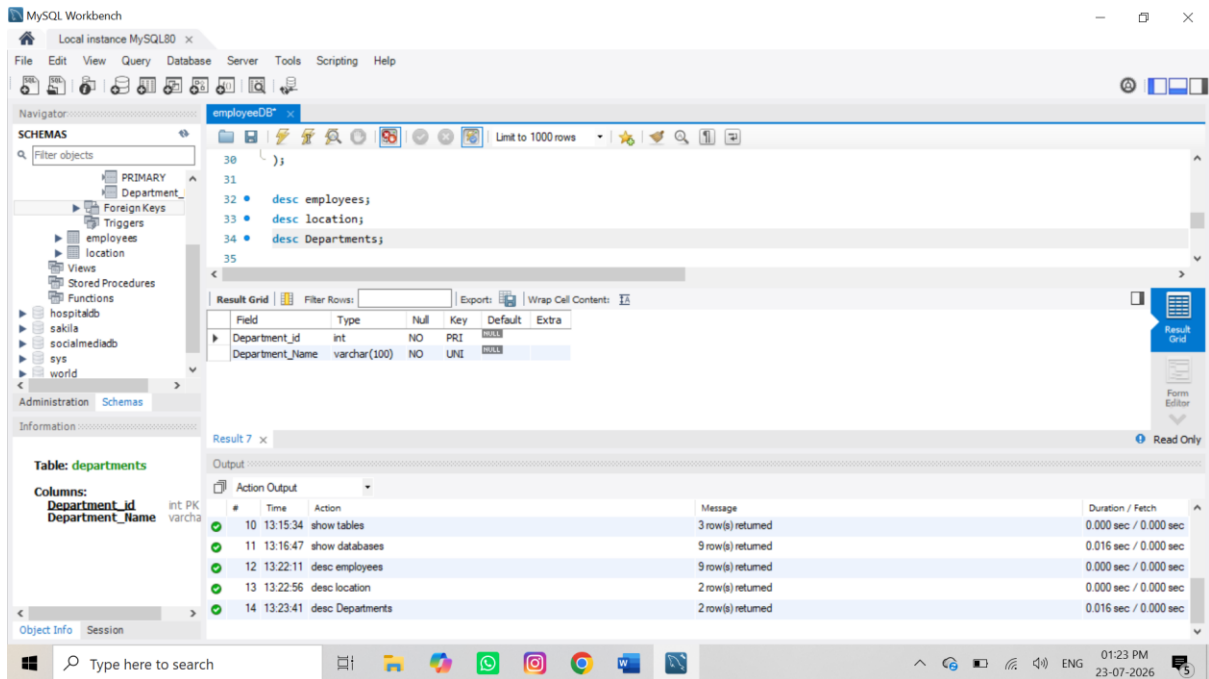
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11	13:16:47	show databases	9 row(s) returned	0.016 sec / 0.000 sec
12	13:22:11	desc employees	9 row(s) returned	0.000 sec / 0.000 sec
13	13:22:56	desc location	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

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2. Table Alteration (ALTER)

- Add a new column named "email" to the Employees table to store employee email addresses.
- Modify the data type of the "designation" column in the Employees table to support a wider range of values.
- Drop the "age" column from the Employees table.
- Rename the "hire_date" column to "date_of_joining".

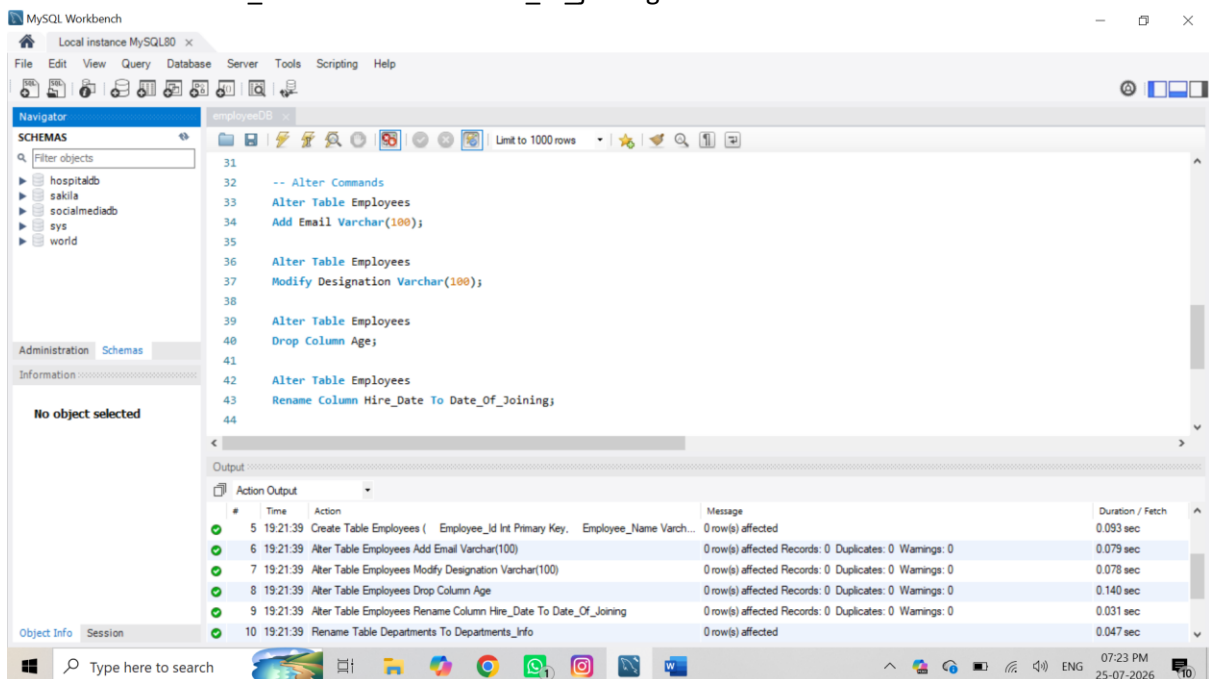
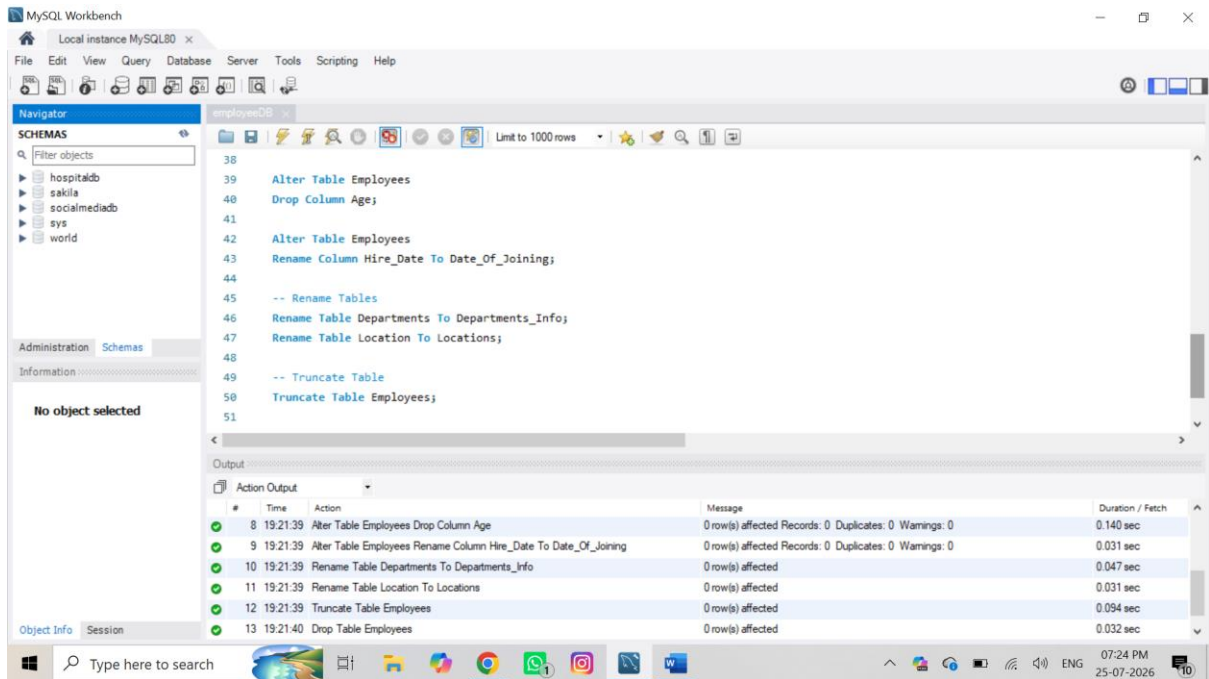


Table Renaming (RENAME):

- Rename the "Departments" table to "Departments_Info".
- Rename the "Location" table to "Locations".

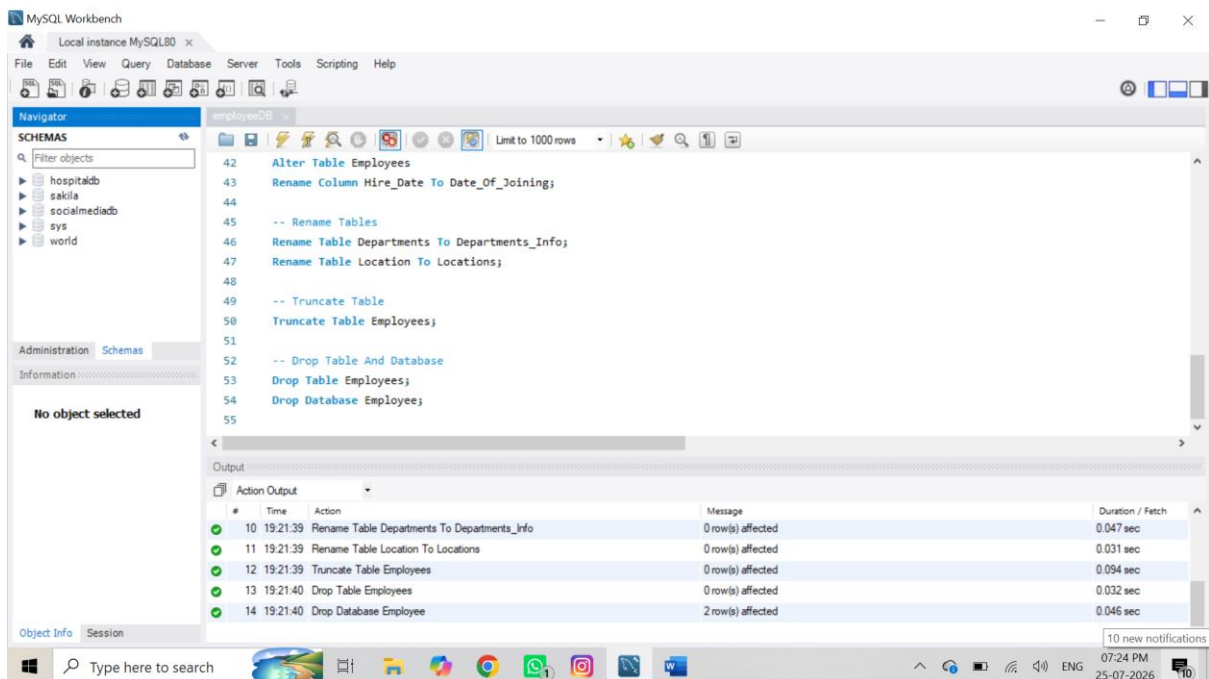


4. Table Truncation (TRUNCATE):

- Truncate the Employees table.

5. Database & Table Dropping (DROP):

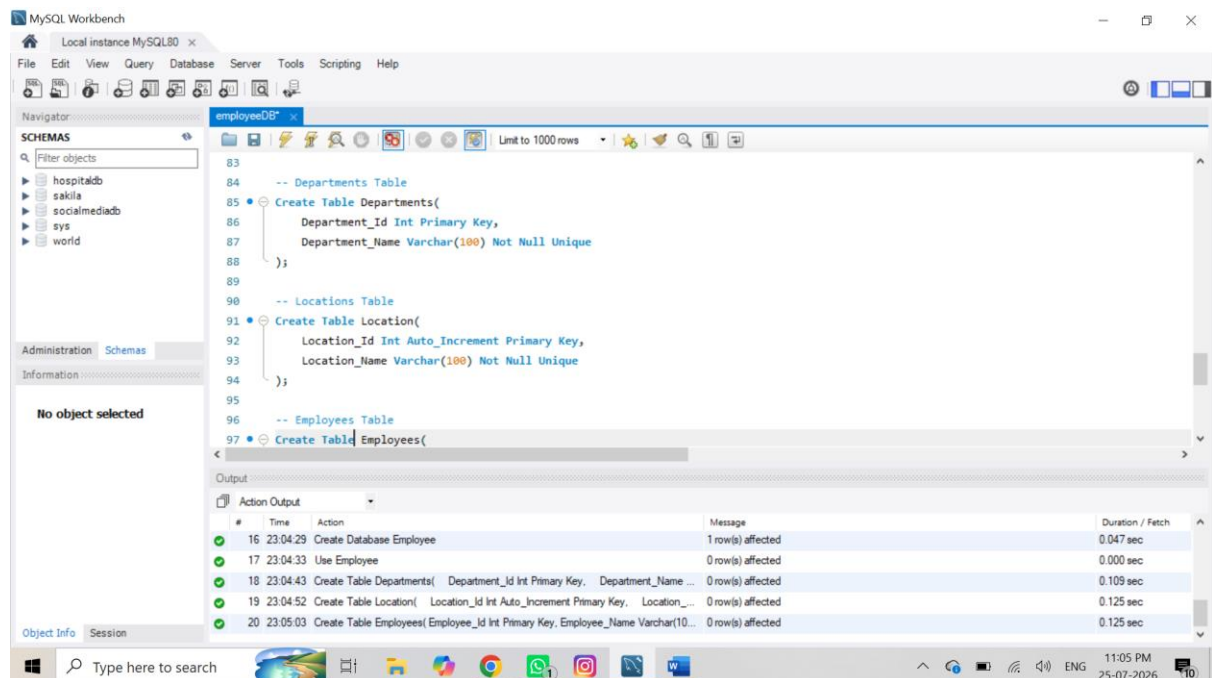
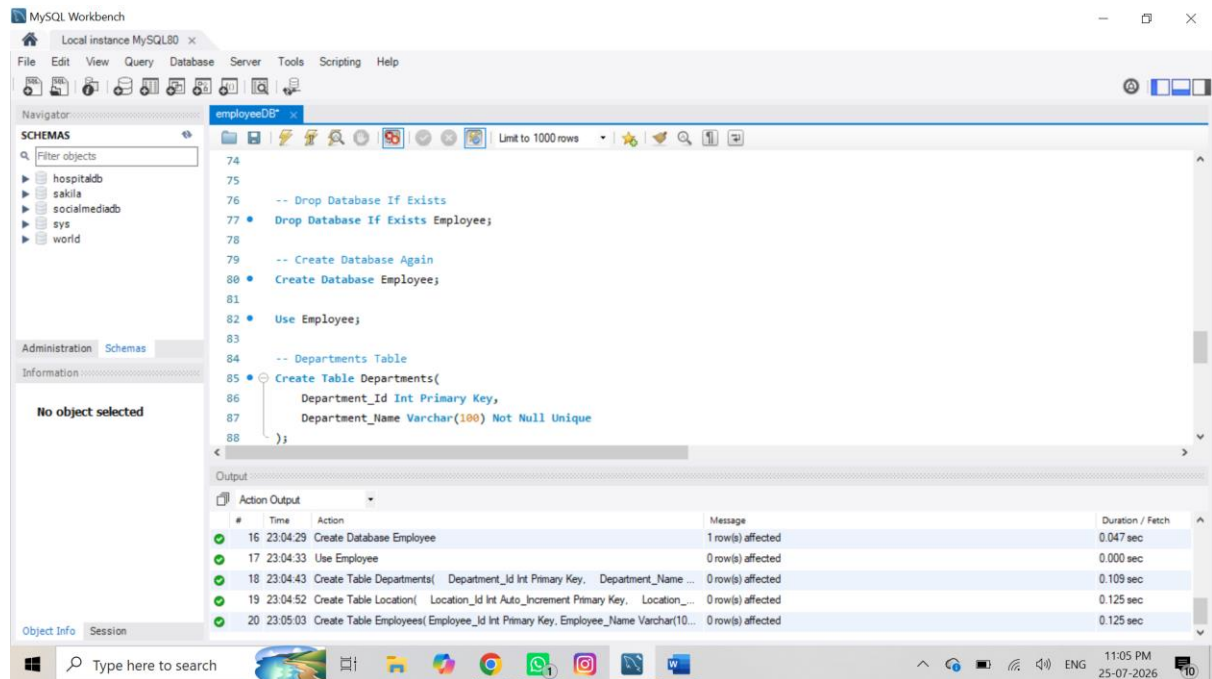
- Drop the Employees table and then the “employee” database.



Tasks: Constraints –

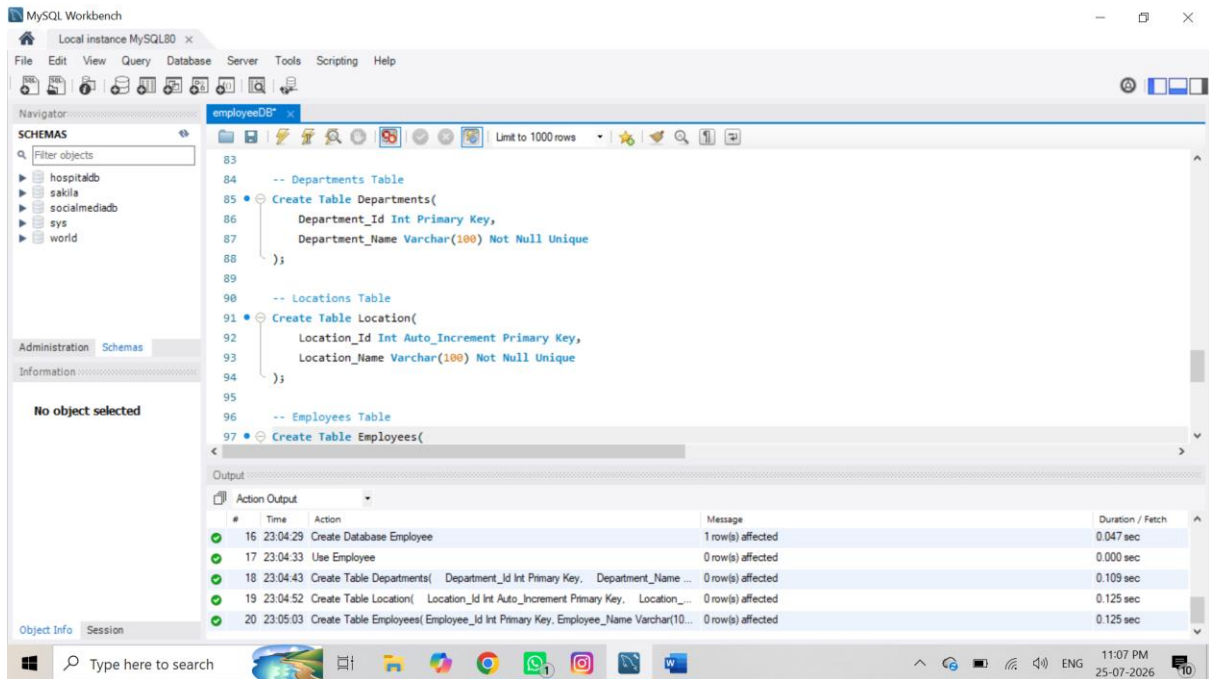
1. Database Recreation:

- Drop the 'employee' database if it exists
- Recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.
- Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.



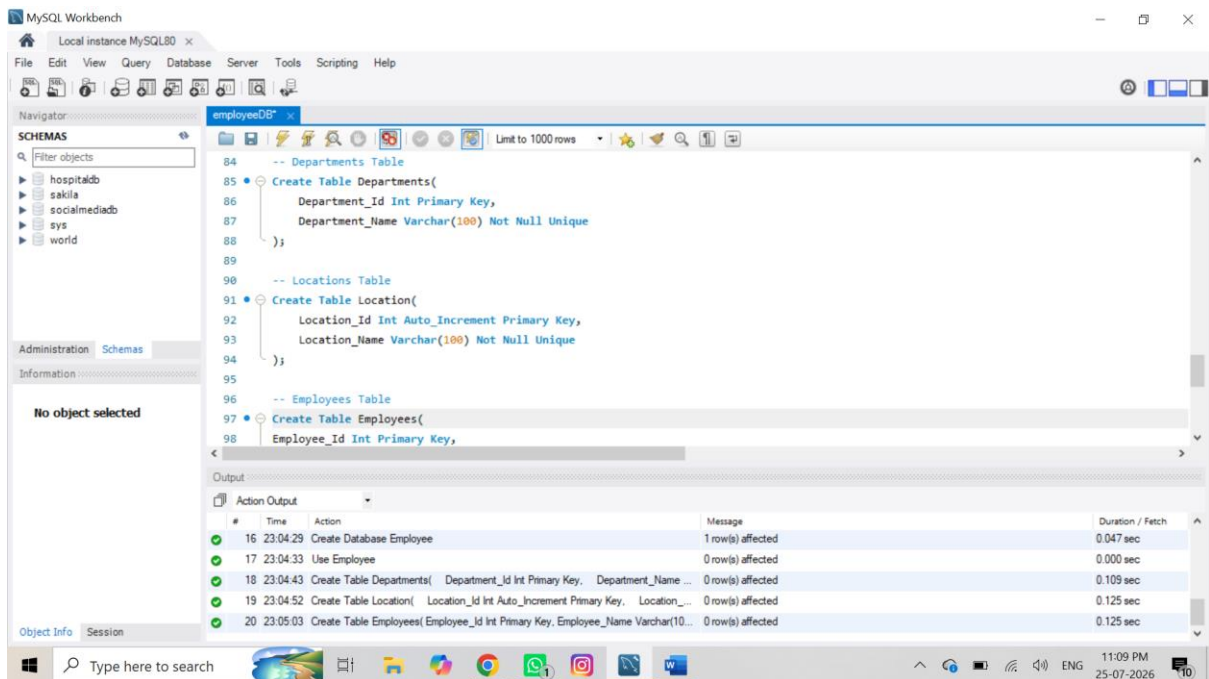
Departments Table:

- Ensure that the "department_id" uniquely identifies each department.
- Set up constraints on the "department_name" to avoid duplicate and null entries.



3. Locations Table:

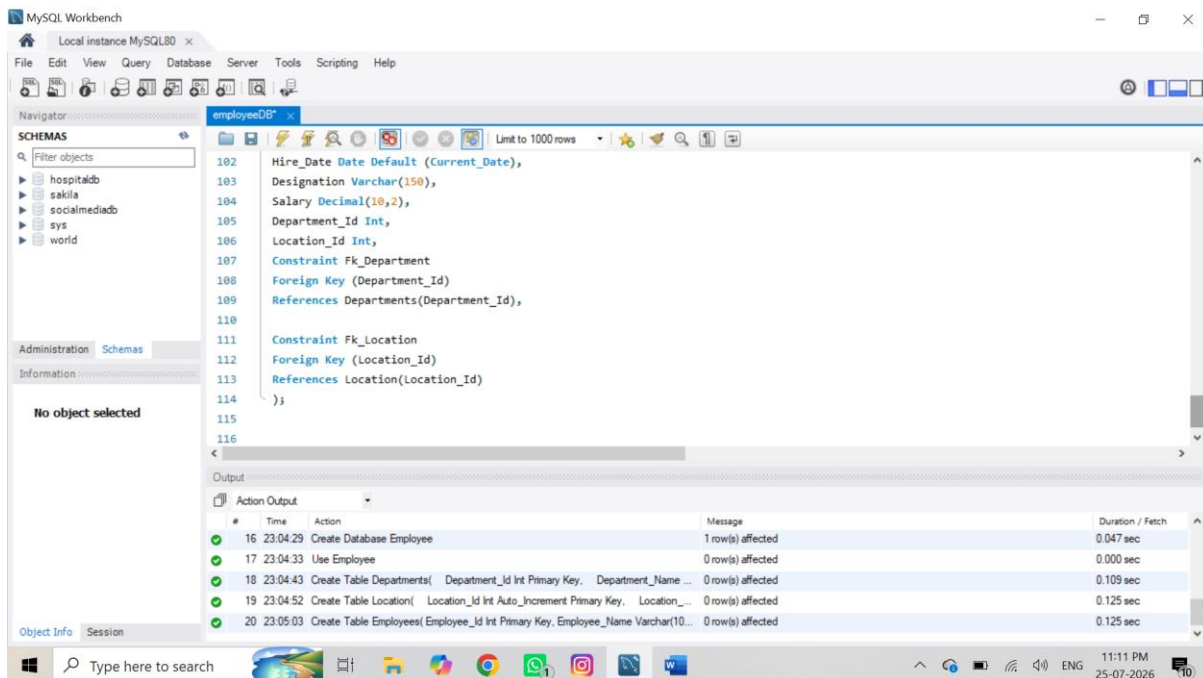
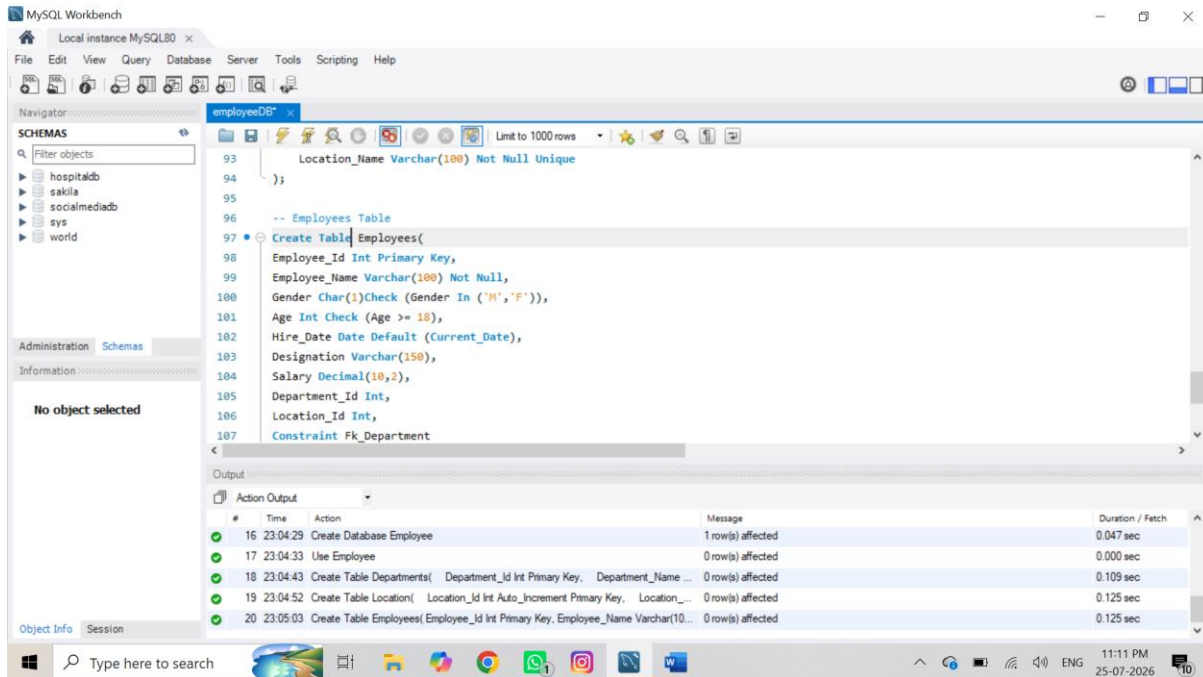
- Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.
- Implement constraints to prevent the insertion of null and duplicate locations.



Employees Table:

- Guarantee that each employee has a distinct identifier.
- Create a restriction to ensure that the employee's name is always provided.
- Limit the acceptable values for the "gender" field to only 'M' or 'F'.

- Enforce a condition to ensure that the employee's age is 18 or above.
- Automatically assign the current date to the "hire_date" field if not specified.



Conclusion

This assignment helped me understand the fundamentals of MySQL Data Definition Language (DDL) commands and database constraints. I successfully created and managed an Employee database by using commands such as **CREATE**, **ALTER**, **RENAME**, **TRUNCATE**, and **DROP**. I also implemented important constraints including **PRIMARY KEY**, **FOREIGN KEY**, **NOT NULL**, **UNIQUE**, **CHECK**, **DEFAULT**, and **AUTO_INCREMENT** to maintain data integrity and establish relationships between tables.

Overall, this assignment improved my practical knowledge of database design, schema management, and relational database concepts using MySQL.